JUNIA Grande école d'ingénieurs

HEI·ISEN·ISA

September–December 2025

ISEN
Module
Numérique Responsable

Lesson 6: Eco design of digital services

TD GROUP

Contrôle continu | 30%
- 40% - Certificat MOOC INR (Institut Numérique Responsable)
- 60% - TD Ecoconception de services numériques

- Group of 4 or 5 person

Max. 11 groups



5 min

Lesson 6 - Today

- Quick Reminder
- Eco design of digital services
- Tools for a web use case
- Examples
- TD Part 1



What is eco-design?

Definition

- **Ecodesign**: Standard for reducing the environmental impact of a product or service.

- **ISO 14062 / IEC 62430:** *"Integration of environmental constraints into the design of products and services using a comprehensive, multi-criteria approach."*

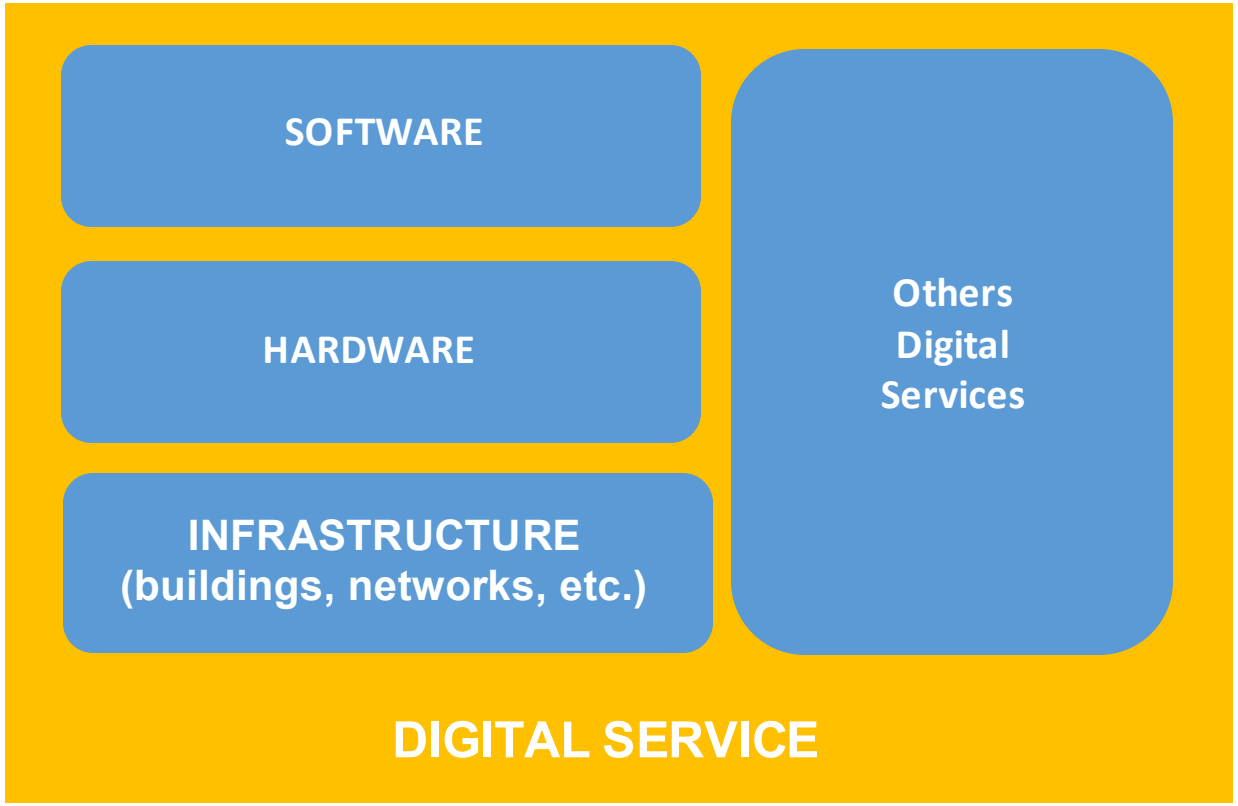
We cannot eco-design a software or a website

We eco-design a business process.



What is a digital service ?

A digital service



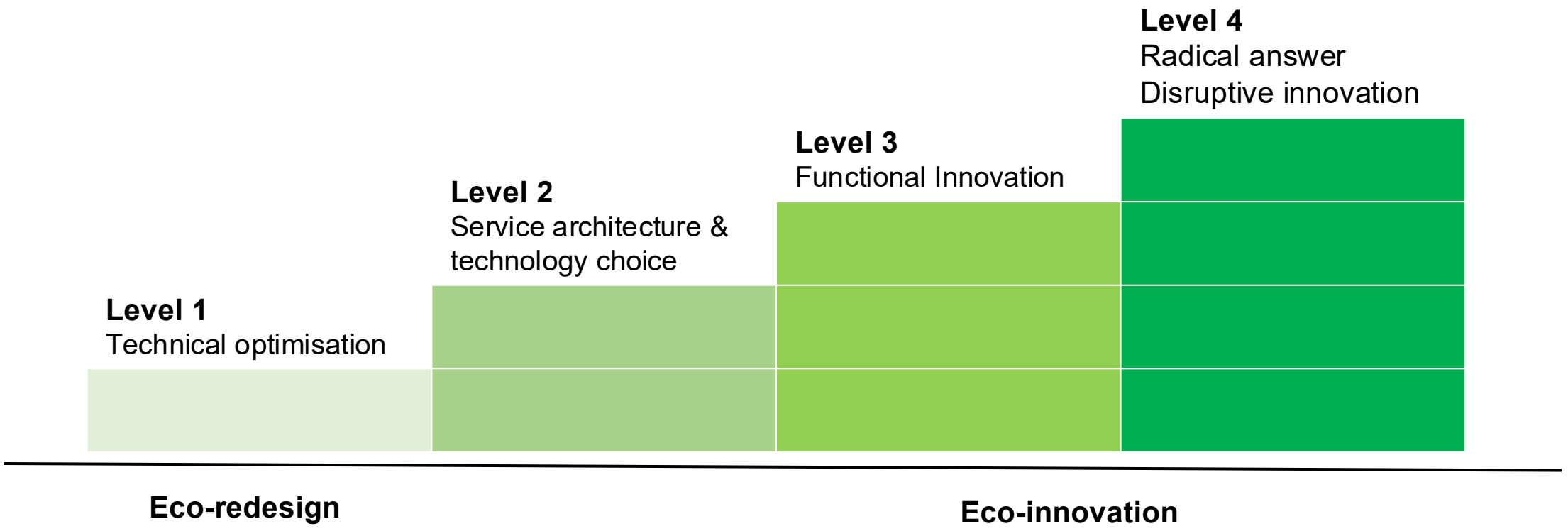
Source: Green IT Collective, From Green IT to Responsible Digital Technology, Glossary, 2018



How many levels up to real eco-design ?

4 Levels of Ecodesign

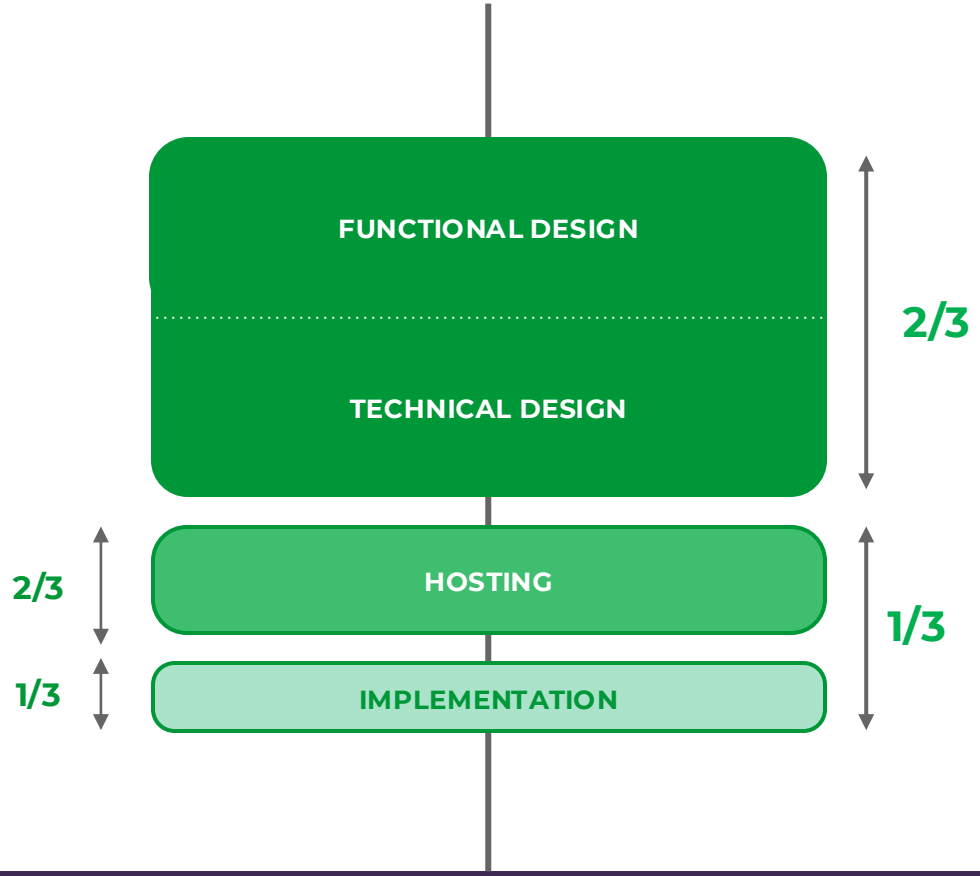
From optimisation to true eco-design



Acting at the level of a digital service

Eco-design approach for digital services

Main impacts in the life cycle



Efficiency and frugality

We seek to reduce digital waste and adopt a more sober approach.

Reducing power

We are seeking to limit the use of IT resources for the operation of the service.

The benefits of eco-design for web sites / app

The benefits of eco-design for websites

- Anticipate Regulations
- Improve user satisfaction
- Improve productivity/conversion
- Reduce in maintenance costs
- Improve SEO (Search Engine Optimization)
- Improve brand awareness and employer branding

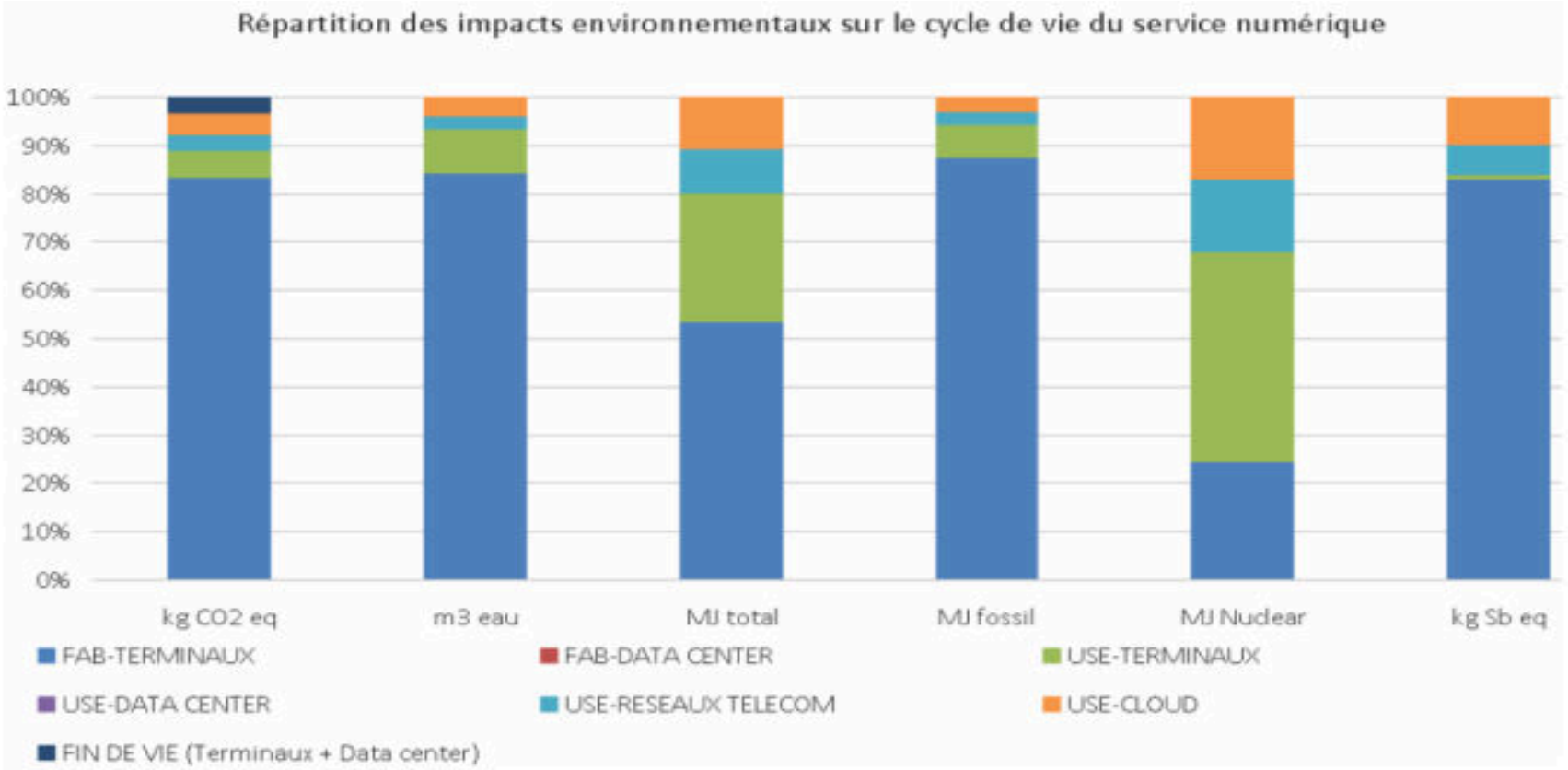
➡️ Promote digital accessibility and quality

.. And beyond the web? *We'll talk about in the examples section.*

The footprint of a digital service

GreenConcept Data

Transactional website



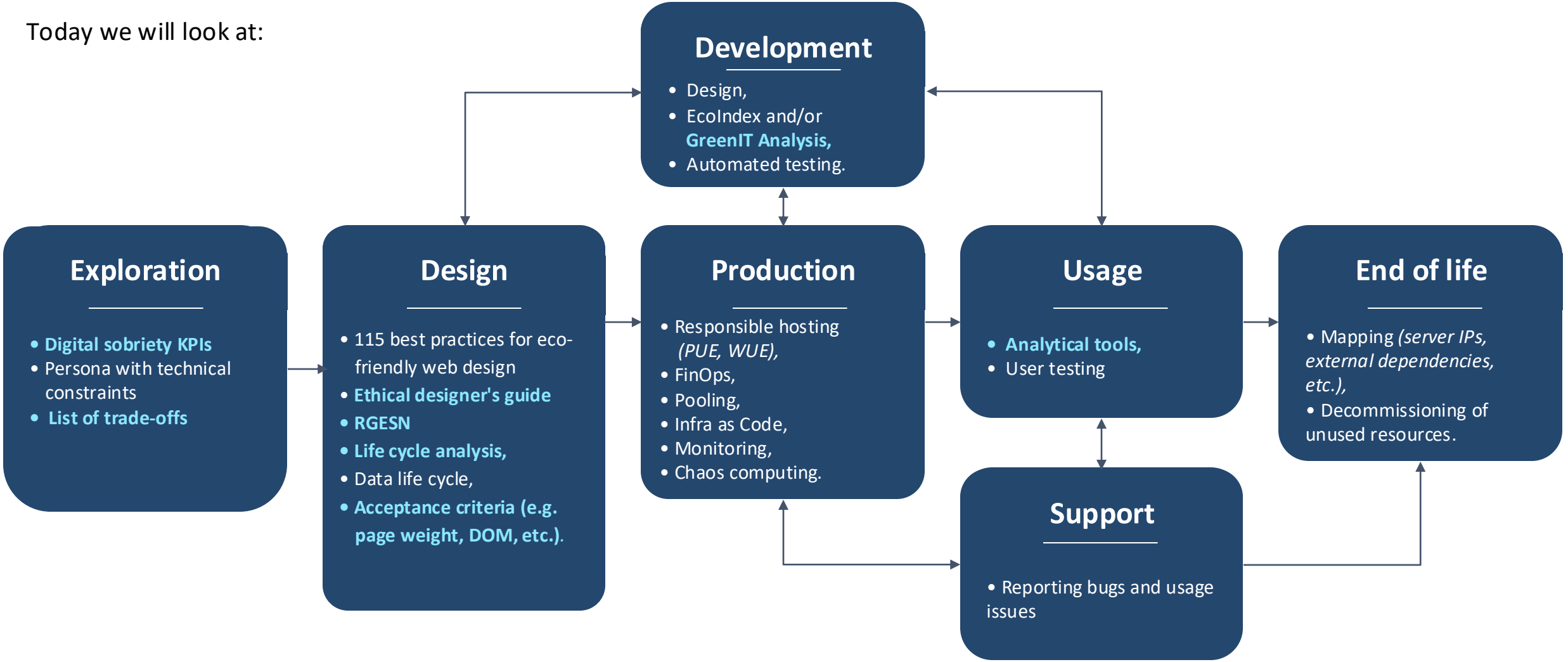
Source: Frédéric Bordage, GreenIT.fr as part of the Greenconcept collective initiative, 2018

Tools

For a web-based approach

Life cycle - Systems and software engineering

Today we will look at:



source: Steps according to ISO 24748-1:2018 standard

Best practice guidelines





115 bonnes pratiques d'écoconception Web

designers éthiques — Guide écoconception

INR — GR-491

Dark patterns guide

When should you digitize?

mi num_ eco — RGESN

afnor — AFNOR SPEC 2201

E-CO — E-commerce white paper

Best practice guidelines for AI

Best practices guidelines



RIA31 - INR



IA frugale - Afnor

Assessment Tools

EcoLogits

Ecologits

ML CO2 Impact

ML CO2 Impact

Further



White book

Assessment support tools

COLLECTIF NUMERIQUE RESPONSABLE

#GIT

eco Index

ADEME — Base empreinte et impacts

green it

ASSOCIATION green it

NEGA OCTET

EcoInfo — Ecodiag

THE SHIFT PROJECT — THE CARBON TRANSITION THINK TANK — Carbonalizer

BOAVIZTA — Données d'impact environnemental

CO2 calculator – impactco2.fr

Ademe



Best practice guidelines
Technical oriented

"Reference" framework

- 115 best practices
- Community work: 40 contributing organisations

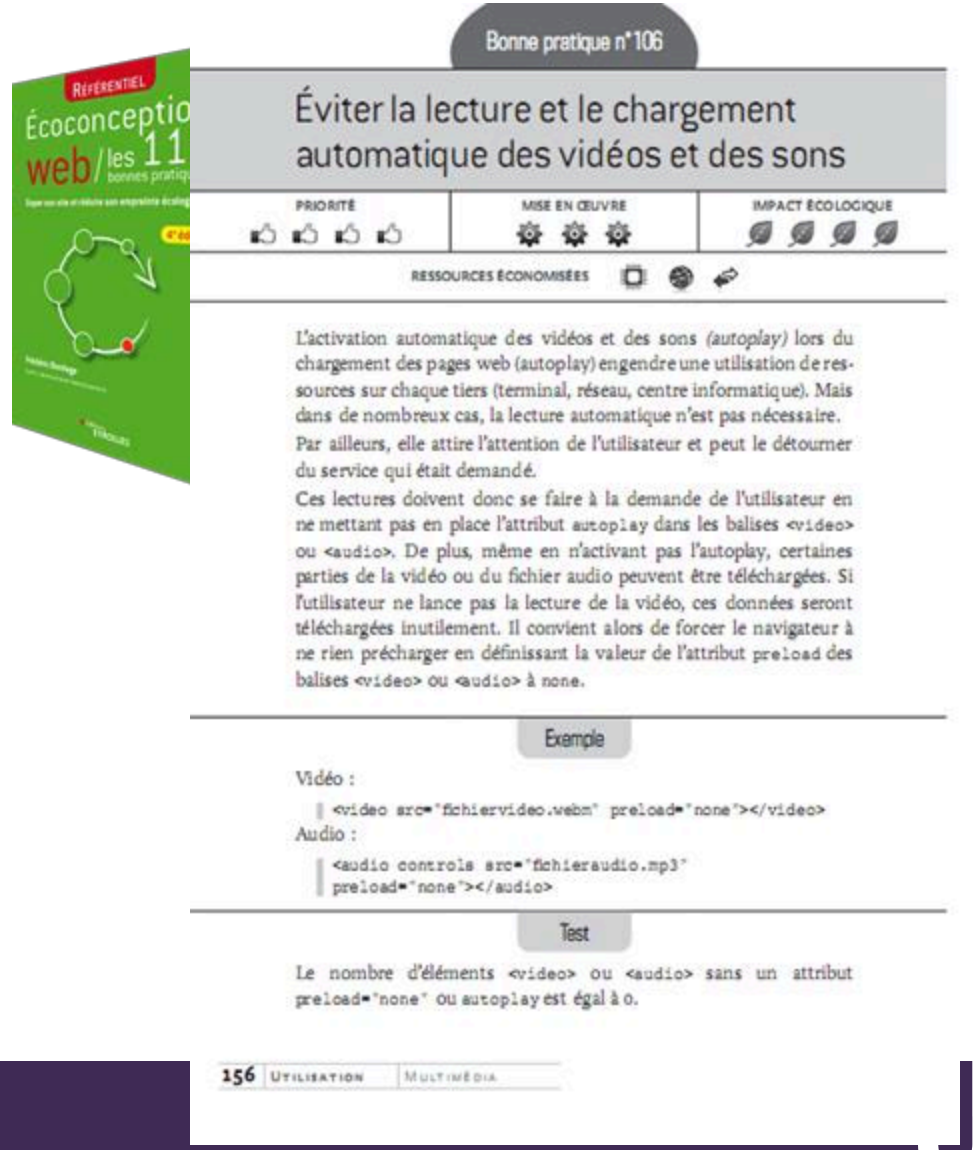
Implementation-oriented

Classified by life cycle stage/profile and priority

- Specification
- Design
- Implementation
- Production (hosting)
- …

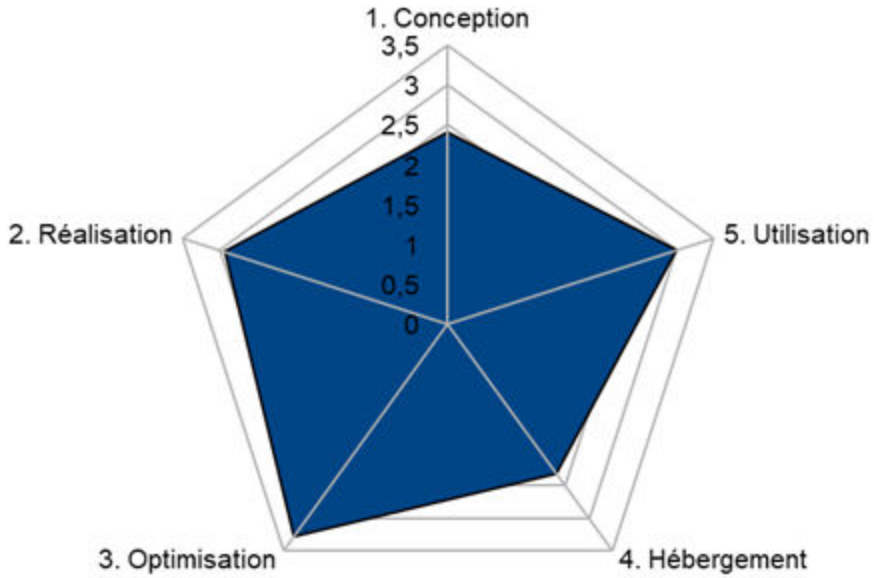
Includes a maturity assessment system

Git Link



Analysis of 115 best practices in web eco-design: sample result

Etape	Catégorie	Moyenne - Note sur 5
1. Conception	1. Fonctionnelle	2,25
	2. Graphique	2,50
	3. Technique	2,45
2. Réalisation	1. Templating	2,18
	2. Code (client)	2,56
	3. Serveur - HTTP	5,00
	4. Serveur - CMS	3,00
	5. Serveur - application	4,00
	6. Serveur - SGBD/R	5,00
3. Optimisation	1. Ressources statiques	3,75
	2. Serveur	2,67
4. Hébergement	1. Fin de vie	0,00
	2. Infrastructure	1,86
	3. Serveur HTTP	3,40
	4. SGBD/R	
5. Utilisation	1. Contenu	3,00
Total Résultat		**2,78**



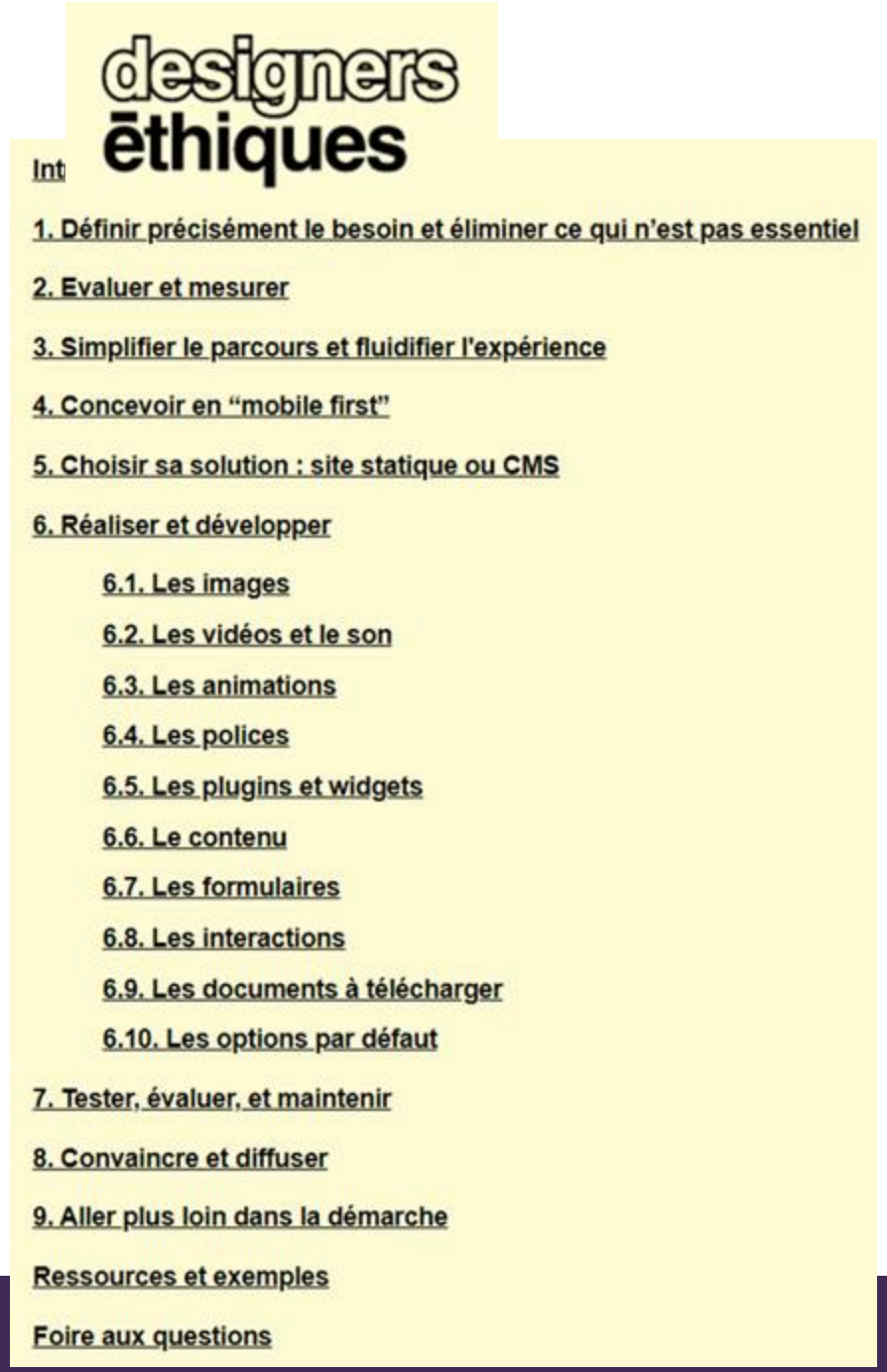
Eco-design guide
Functional & UX oriented

Action research association focused on responsible and sustainable design

The guide includes:

- Explanations of best practices
- Questions to ask yourself during the scoping and implementation phases
- Examples of things to do and not to do

The guide (fr)



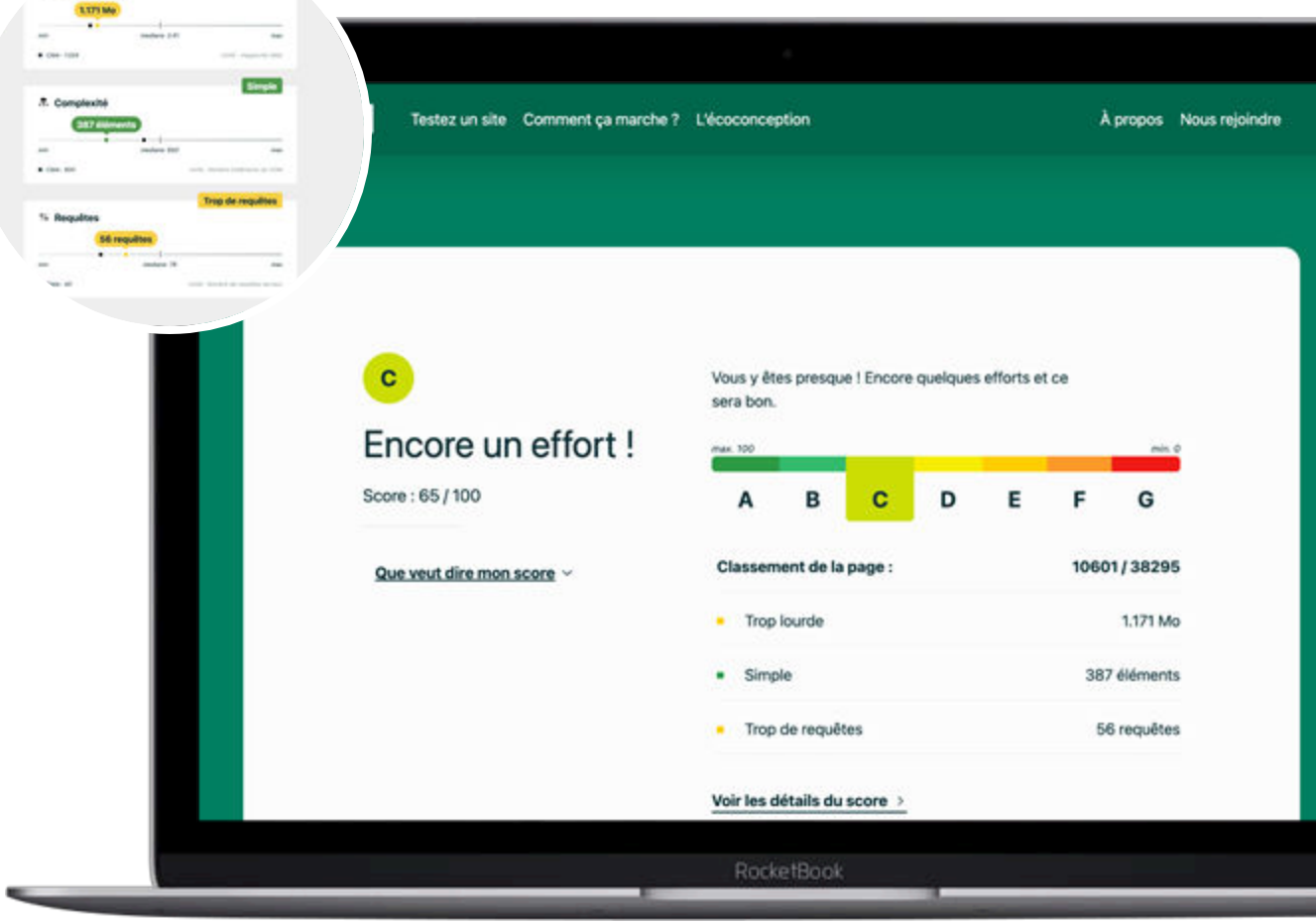
EcoIndex
Part of next TD!

- Free, transparent, open algorithm
- The benchmark for professionals (tens of thousands of users)
- Numerous large companies (EDF, Société Générale, Publicis, etc.)

For a URL:

- environmental performance;
- technical footprint of the page (weight, complexity, etc.);
- associated environmental footprint (greenhouse gases
  and water).

[EcoIndex.fr]



EcoIndex and Target Calculation



EcoIndex calculation: $100 - \frac{5\,(3QtilDOM + 2QtilHttp + QtilKo)}{6}$

Parameters	Coefficient	Median	Target - Grade C	Impacts
Number of DOM elements	3	690	600	Terminal CPU, RAM
Number of HTTP requests	2	78	40	Server CPU, RAM
Page size	1	2200 KB	1000 KB	Network and terminal RAM

GreenIT Analysis Plugin

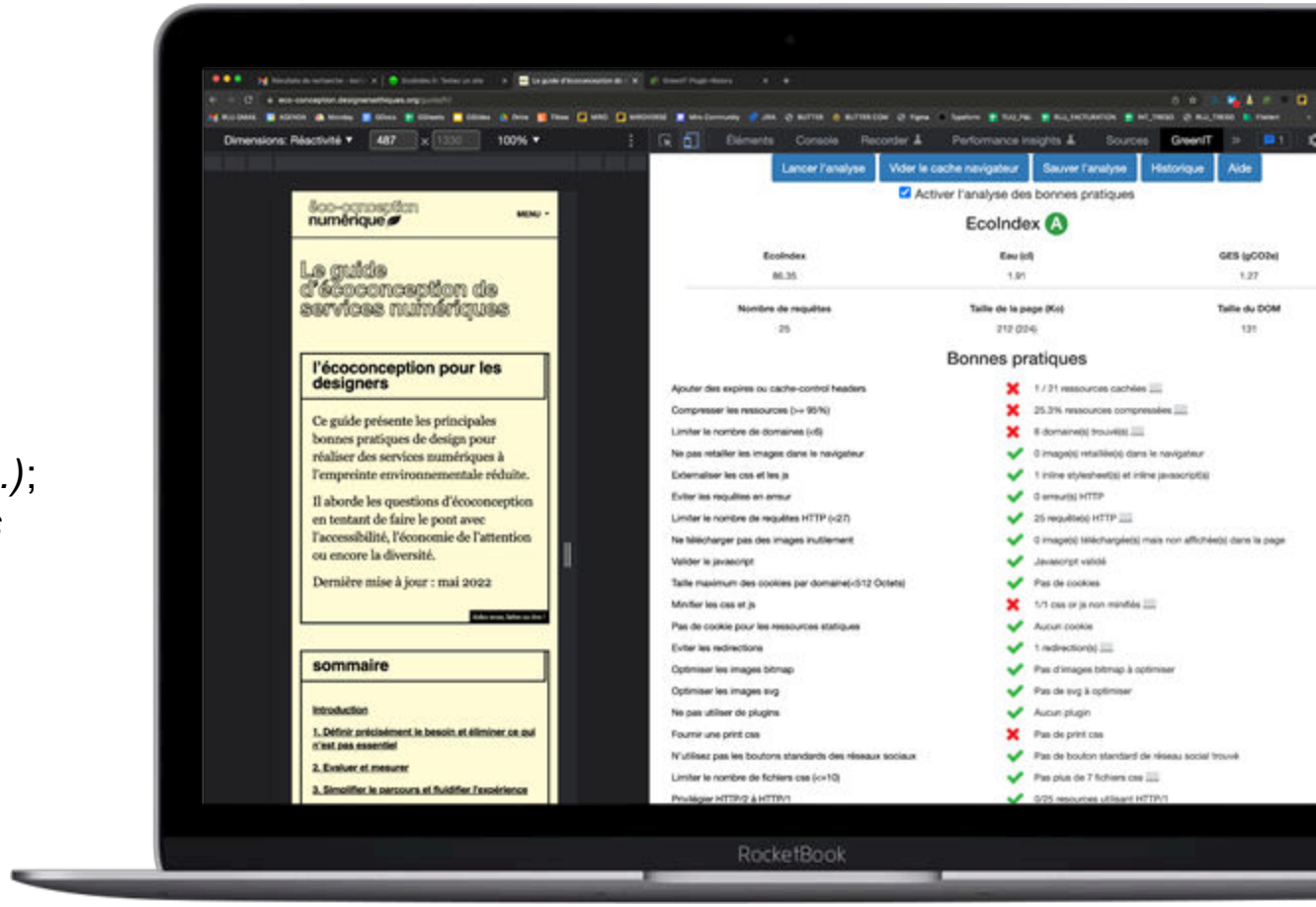
- A free, transparent, open community tool
- Based on EcoIndex and the web eco-design reference framework,
- Reference tool available in various formats:
  - browser extension *(Chrome, Firefox),*
  - CLI.

For a URL:

- Environmental performance
- Technical footprint of the page *(weight, complexity, etc.)*;
- Associated environmental footprint *(greenhouse gases and water).*

Firefox

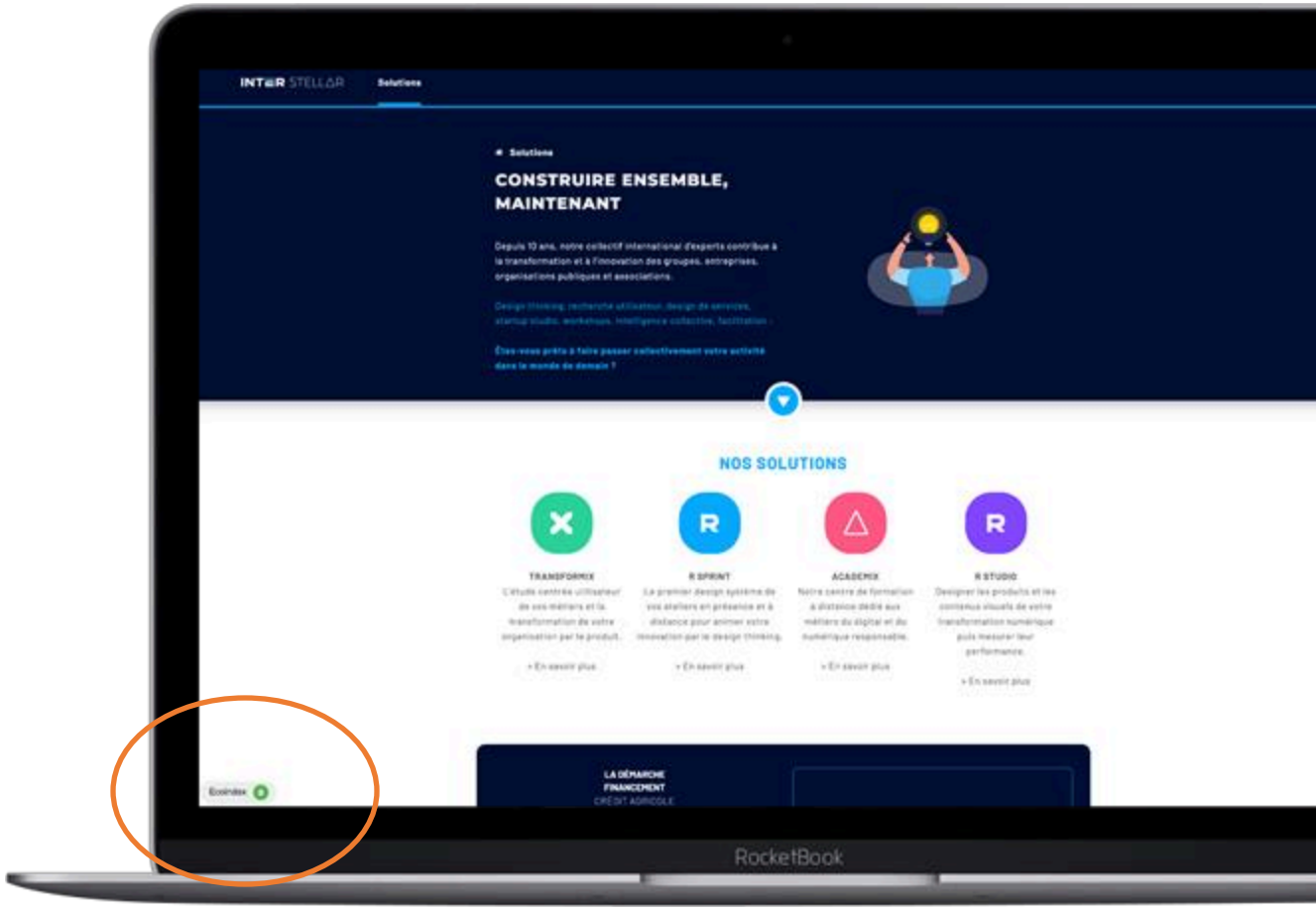
Chrome



EcoIndex - the badge

The aim of this project is to provide a badge that can be easily integrated into your web pages. This badge will display the Ecoindex score of the web page on which it is integrated and will redirect the user to the Ecoindex score details page.

Git Link



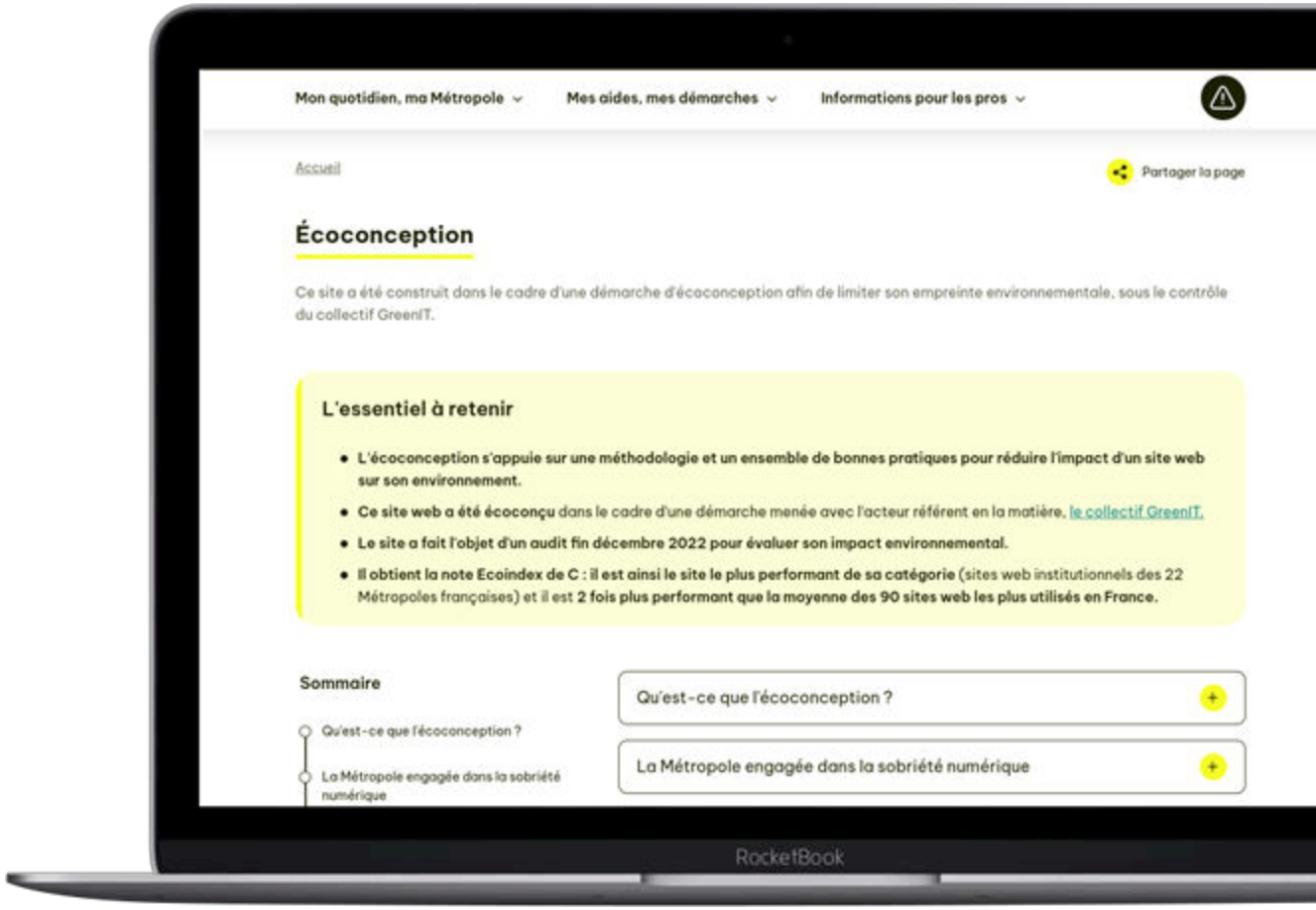
https://interstellar.eco/

The environmental statement

A community tool that highlights the work done on eco-design

Generator

Example



How to evaluate?

Estimation exercise with Ecoindex

Steps to evaluate

1. Define scope
2. Functional Analysis
3. Technical Analysis
4. Quantification of impacts
5. Recommendations

1/ Scope

- Define the digital service being used
- Define the functional unit(s)
- Identify the stages of the journey
- Identify the persona

1/ Scope

The functional unit : What is it ?



> Reminder / Lesson 4

1/ Scope

The functional unit

What is it?

The **main function** of a product or service:
a representative **reference unit**, corresponding to a business activity.

What is it for?

- **Assessing** the environmental footprint of a business activity.
- **Compare** services or products when conducting a Life Cycle Assessment (LCA).
- **Reflect** on essential needs.

Examples of functional units

- Buying a concert ticket
- Looking up a telephone number
- Make an appointment
- Checking your bank account
- Watch a video (for 10 minutes)

A service may have several functional units, but their number **must remain limited.**

2/ Functional analysis - Eliminate non-essential features

Using the C.O.E.U.R method to reduce complexity and focus on the essentials

- **C for Conceal :** Hide non-essential interface elements from view.
- **O for Organise:** Bring hierarchy and structure to the interface by organising certain components differently.
- **E for Eliminate:** Simply remove a component from the screen.
- **U for Uniformise:** Standardise and harmonise elements with each other.
- **R for Reduce:** Reduce the importance of an interface element.

Nb : COEUR means HEART in french

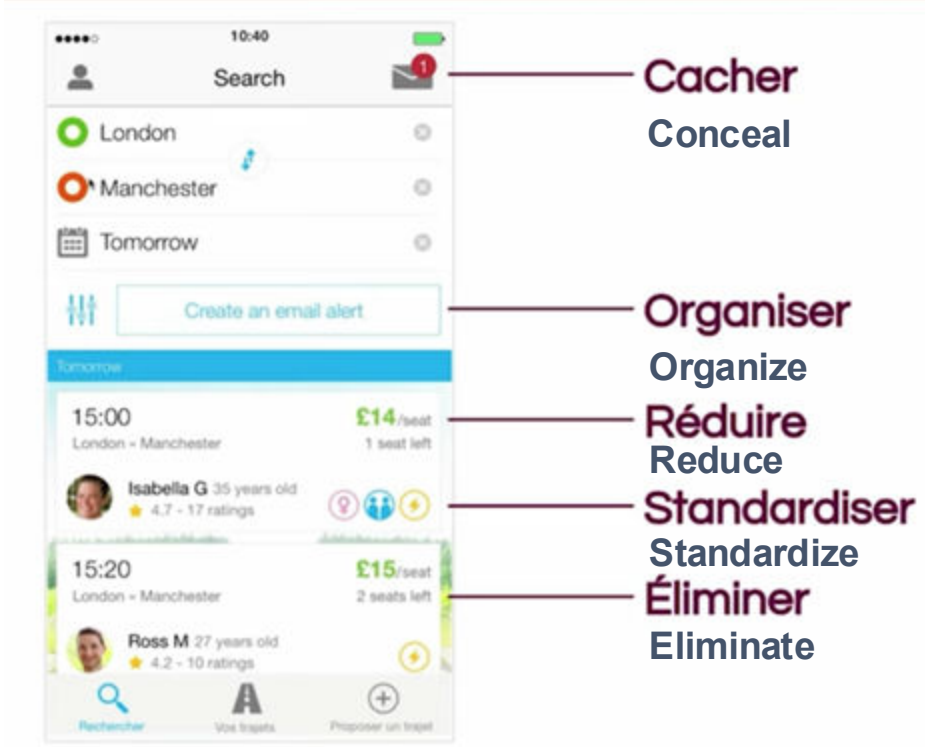
Source: https://www.lunaweb.fr/actualites/blog/atelier-ideation-eco-conception/

2/ Functional analysis - Eliminate non-essential features

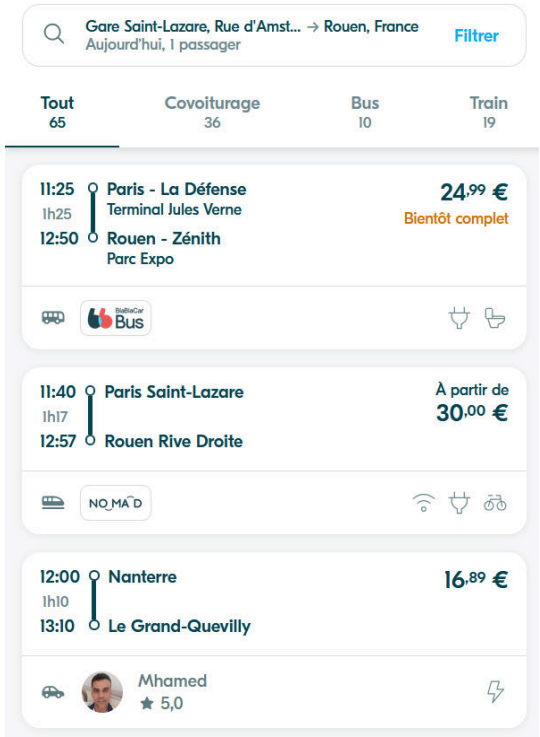
Example: BlaBlaCar

Using the C.O.E.U.R method to reduce complexity and get to the heart of the matter

Before



After

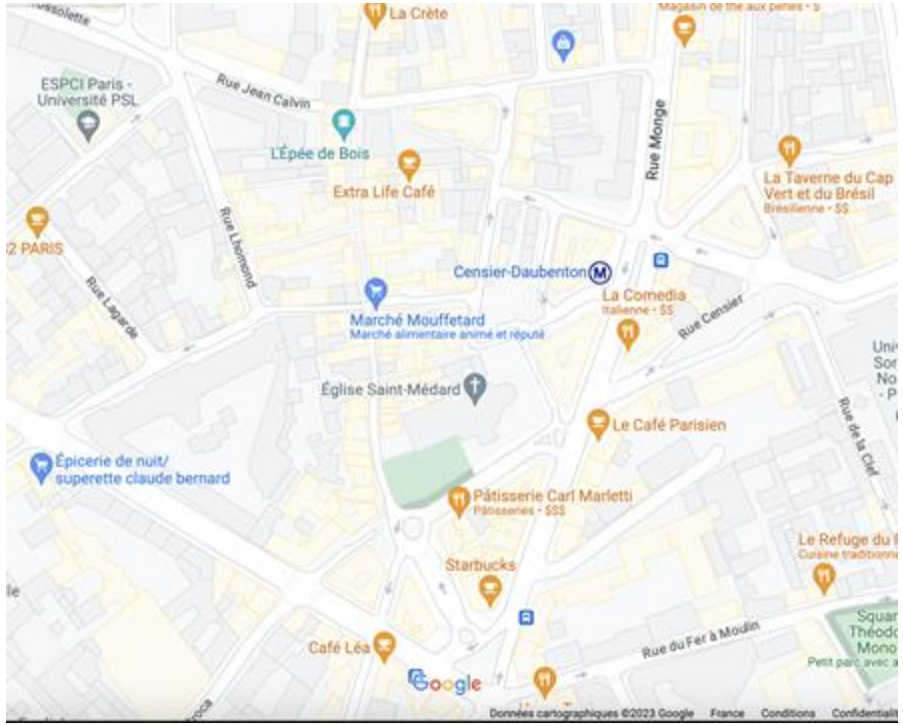


"Combating complexity" - Rémi Guyot (video)

2/ Functional analysis – Meeting the right user needs

Example of a feature that does not meet an essential user need

Google Maps: advertising revenue vs. usability



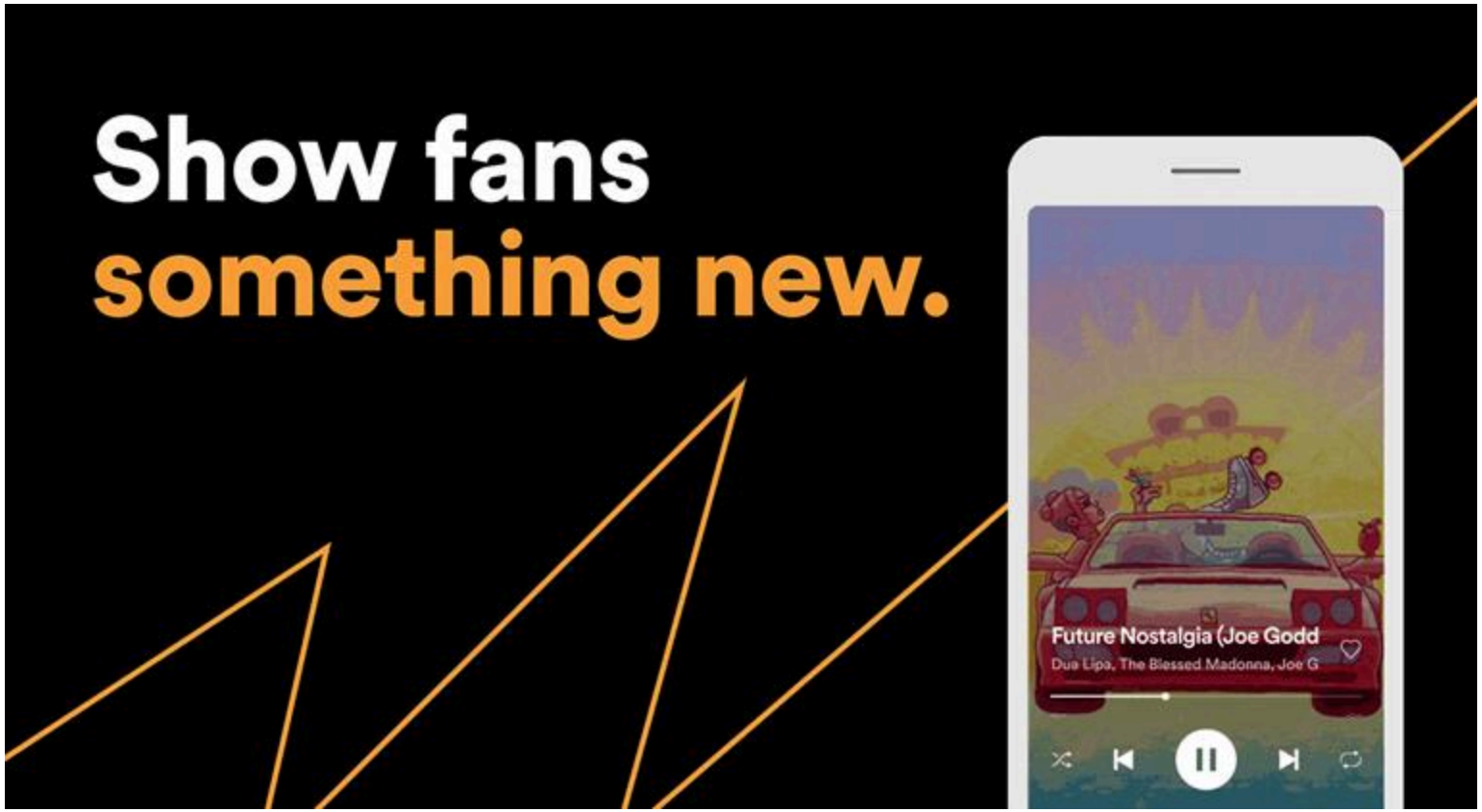


Open Street Map: highlighting street names, pedestrian streets, public buildings, bus stops, etc.

2/ Functional analysis – Meeting the right user need

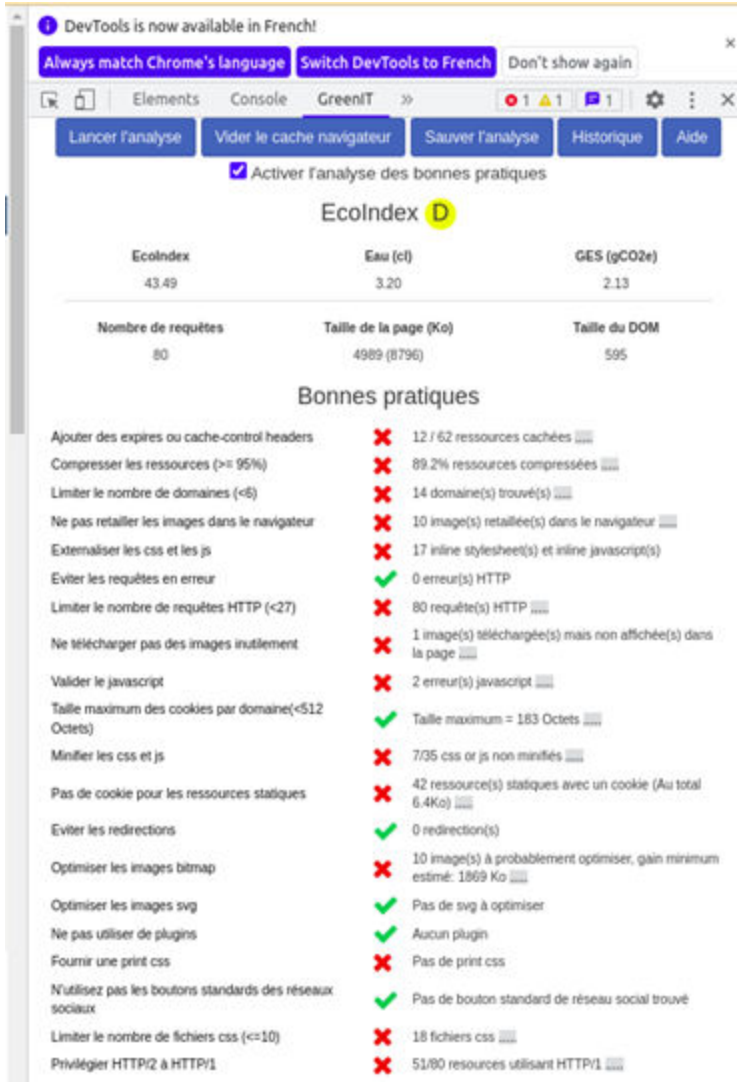
Example of a feature that does not meet an essential user need

Spotify Canvas: capturing attention vs. inclusion



3/ Technical analysis

- Comparison of page weight/number of requests/DOM with an EcoIndex A
- Analysis of 115 best practices relevant to the project
- Can be supplemented with other tools such as Lighthouse



4/ Quantification of impacts

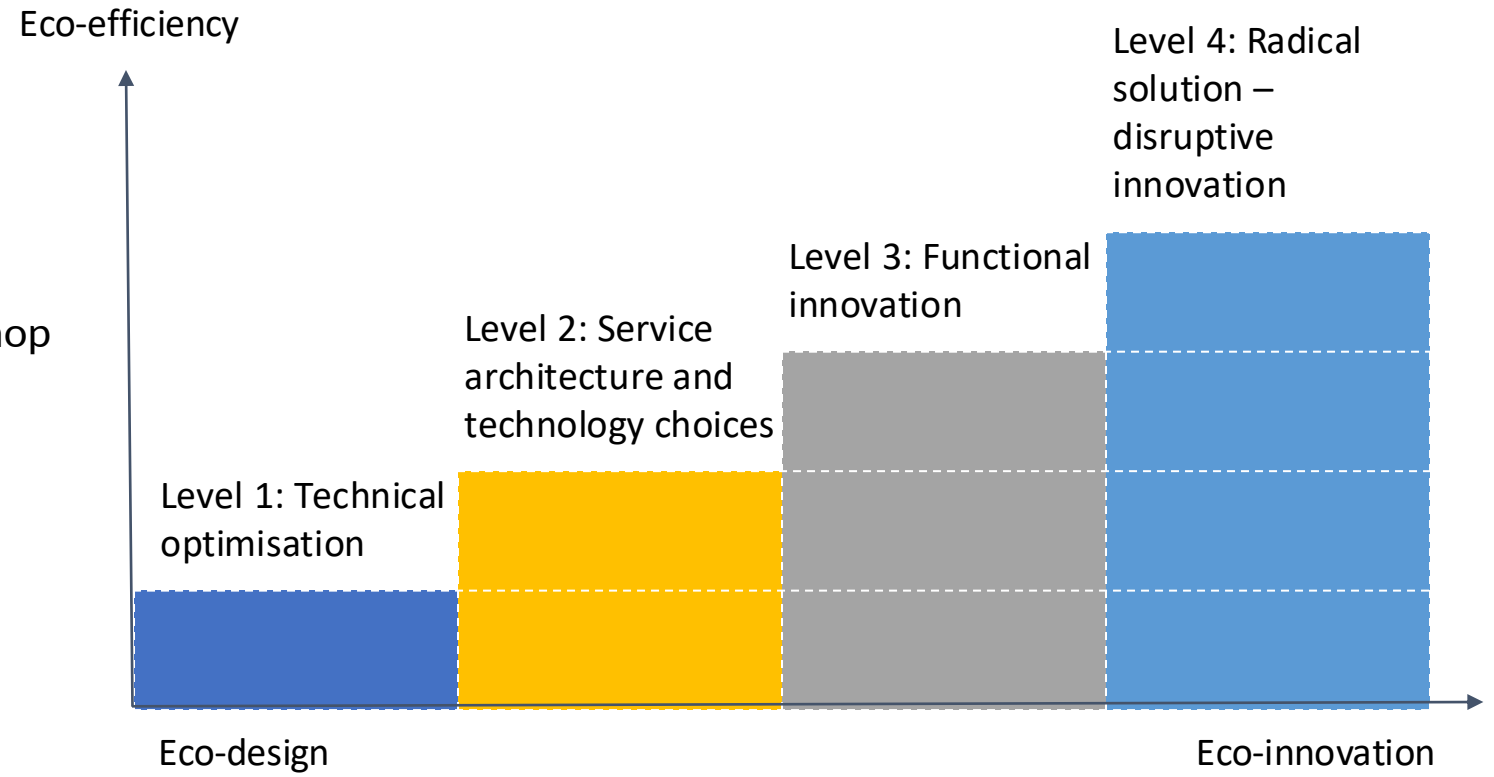
- Export of environmental performance for the functional unit
- Conversion into meaningful equivalents:
  - Shower, bath, pack of water
  - Kilometres by car, distance between Lille and Paris, etc.

Url	Request number	Size(kb)	Dom size	Greenhouse Gases Emission (gCO2e)	Water consumption (cl)	EcoIndex	Grade
https://www.impots.gouv.fr/portail/	45	694	806	2	3	50	D
https://www.impots.gouv.fr/portail/recherche/je%2520d%25C3%25A5	103	995	552	2.12	3.18	44	D
https://www.impots.gouv.fr/portail/particulier/payer-mes-impots-taxe	32	341	382	1.6	2.4	70	B
https://www.impots.gouv.fr/portail/particulier	39	562	654	1.86	2.79	57	C
https://www.impots.gouv.fr/portail/particulier/je-consulte-mes-informa	39	581	367	1.64	2.46	68	B
https://www.impots.gouv.fr/portail/particulier/gerer-mon-prelevemen	32	270	390	1.6	2.4	70	B
https://www.impots.gouv.fr/portail/particulier	41	423	654	1.86	2.79	57	C
https://www.impots.gouv.fr/portail/recherche/Restitution%2520de%25	36	189	466	1.68	2.52	66	B
https://www.impots.gouv.fr/portail/particulier	35	287	654	1.82	2.73	59	C
https://www.impots.gouv.fr/portail/	51	1046	806	2.06	3.09	47	D
			Total	12.52	27.36	58.8	
			x 1 000 000				
				12520000g	27360000cl		
				12.52 tonnes	273 600 litres		

5/ Recommendations

- Consolidate Level 1 and 2 recommendations
- Organise a brainstorming workshop to move towards Levels 3 and 5
  - Ideation
  - Tool for determining the externalities of a project





Source: Eco-design cluster, https://www.eco-conception.fr/library/h/note-de-synthese-numerique-version-abregee.html

A few examples by levels

Not only web oriented

Technical – Factor 1
Retail

🔋 Battery life

Reduce the refresh rate of electronic label prices to extend their lifespan

➡️ Work on possible optimisations to reduce the number of refreshes on electronic label screens (the most energy-consuming element).
Thanks to this project, **a two-year increase in lifespan** has been achieved.
From an environmental perspective, this equates **to a 28% gain over the entire lifespan of the equipment** for this major retailer.



Functional – Factor 2
Retail

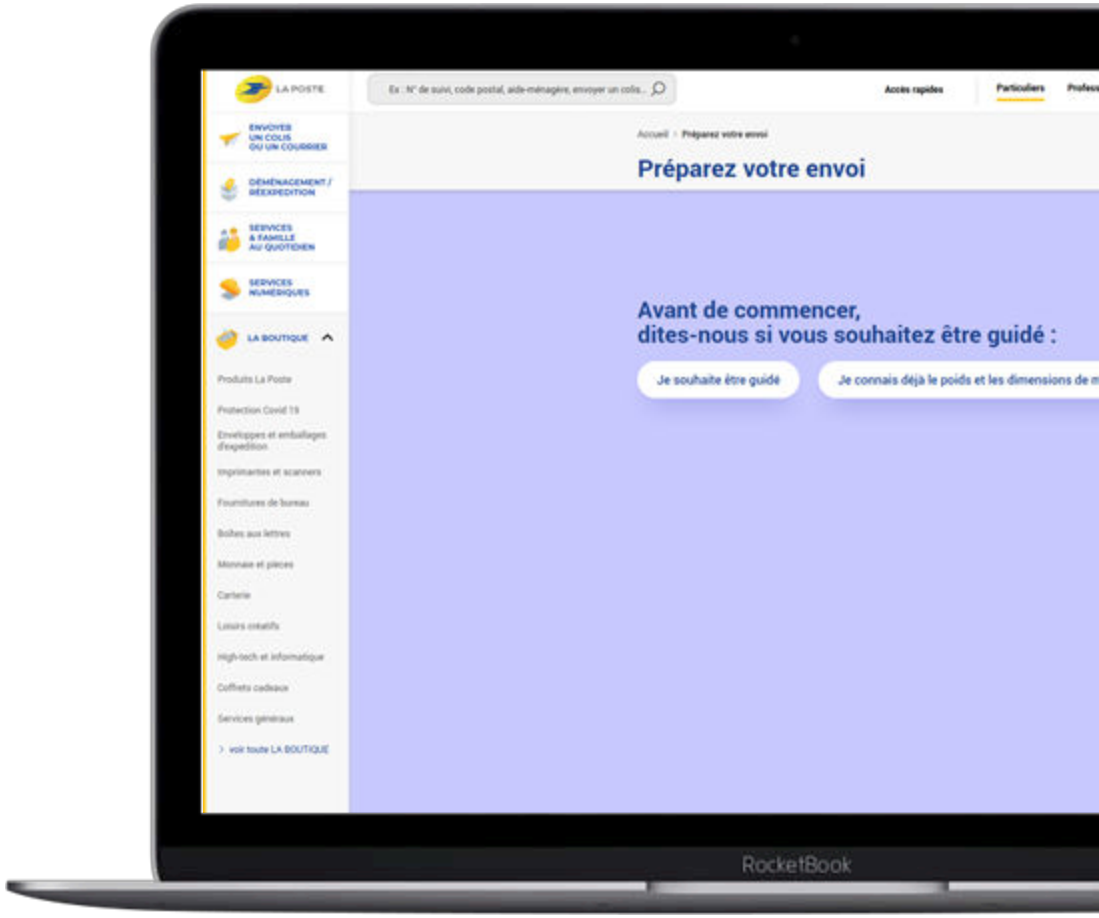
BEST PRACTICE

Inform the user that they must have a printer.

FOOTPRINT

The entire footprint of the functional unit is avoided if the user does not have a printer.

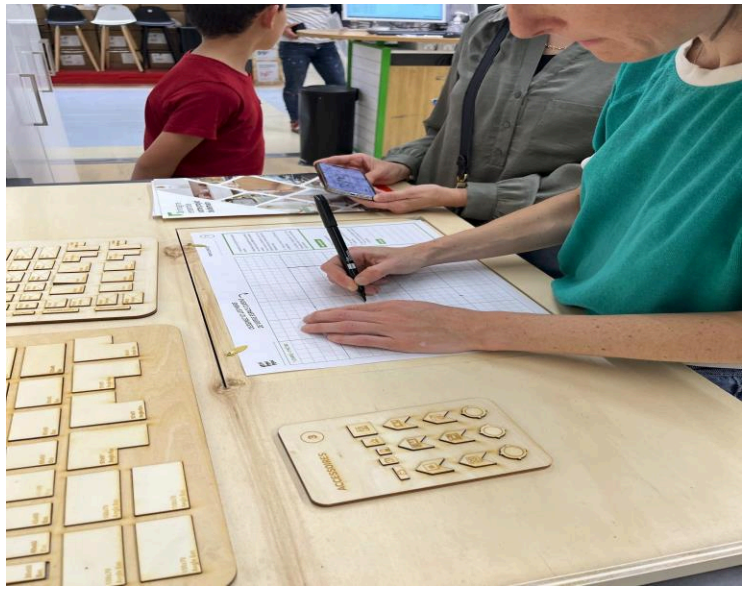


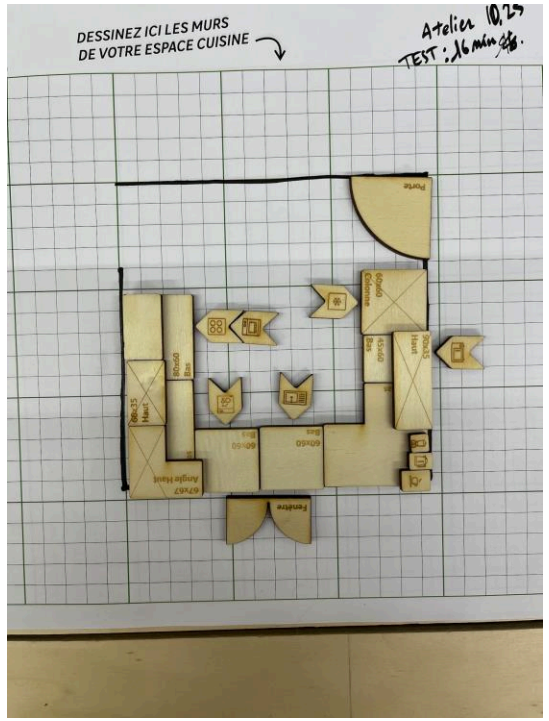


Functional – Factor 3

Retail

Modelling a kitchen project by hand, prior to a more precise configuration
with a 3D simulator, is more interactive and allows you to better understand the customer's needs quickly



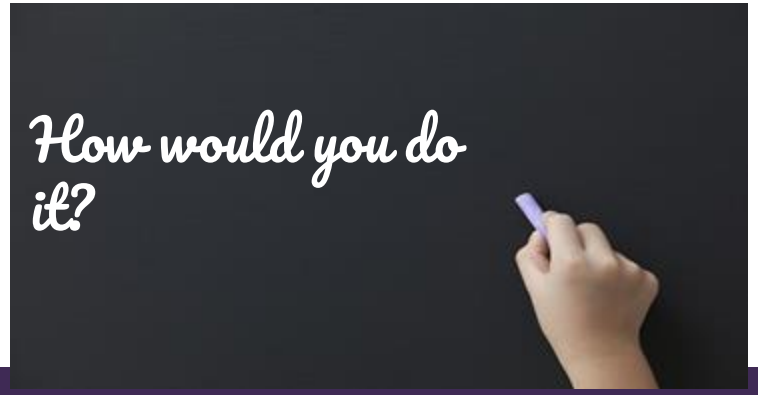


#Low tech # Fun

Functional - Factor 3 - 4

Weatherforce





Tutorial > what to do, step by step

Duration: 1 hour

1. **Framing the audit - 10 min**
2. **Functional analysis - 15 min**
3. **Quantify the impact - 20 min**
4. **Technical analysis – 15 min**
5. **Recommendations**

NEXT CLASS > Presentation to convince about eco-design

1. **Framing the audit - 10 min**
    a. Defining the functional unit
    b. Define a specific scenario (path)

2. **Functional analysis - 15 min**
    a. How much time spent on each page
    b. Complexity of understanding the content and doing what you want to do

3. **Quantify the impact - 20 min**
    a. Analysis of each page of the journey with GreenITAnalysis
    b. Export the entire journey in CSV format and import into a spreadsheet
    c. Convert GHG emissions into equivalents (km travelled by car, return flight, carbon footprint of a French resident, etc.) - Excel calculator
    d. Convert water into equivalents (cans, showers, swimming pools, etc.)
    e. Estimate over 1 year based on traffic (https://www.similarweb.com/fr/)
    f. Potential savings

4. Technical analysis - 15 minutes

a. Analysis with recommendations GreenITAnalysis + complementary tools
b. Comparison with the EcoIndex target

5. Recommendations F1 to F4

a. F1
    i. Gzip compression of requests
    ii. Caching header with high retention
    iii. Image size and weight
    iv. SEO improvement

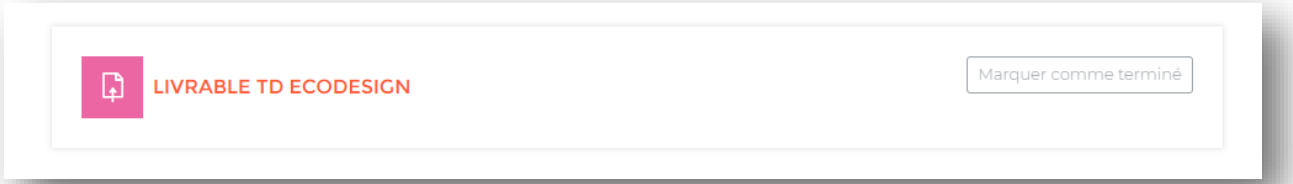
EXAMPLES

b. F4
    i. Is this the right technology?
    ii. Is it possible to use low-tech or no-tech solutions?
    iii. Estimated gain? Divide by 4

6. Presentation with arguments to convince stakeholders to adopt eco-design: next class!

Practical Work Deliverable

- Upload your work on junia learning at the end of the class
- Document Name: NAME1-NAME2.ext
- List of all students in the document



Next class : PRESENTATION
8 minutes per group
(5 minutes presentation + 3 minutes Q&A)

Evaluation Criteria

Expression / Support quality	3
Steps compliance	2
Synthesis capacity	3
Content relevance	8
Answers relevance (Q/A)	4

JUNIA Grande école d'ingénieurs

HEI·ISEN·ISA

September–December 2025

ISEN
Module
Numérique Responsable